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HRB 22.2V 4000mAh 60C XT60 Lipo Battery

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HRB 6S Lipo Battery User Manual

Model: 22.2V 4000mAh 60C XT60 Lipo Battery

1. Introduction

Thank you for choosing the HRB 6S Lipo Battery. This manual provides essential information for the safe and effective use, maintenance, and storage of your new lithium polymer batteries. Please read this manual thoroughly before using the product to ensure optimal performance and safety.

2. Product Overview and Features

The HRB 6S Lipo Battery is designed for high-performance RC applications, offering reliable power and a long cycle life. Key features include:

- **High Capacity:** 4000mAh for extended running times.
- **High Discharge Rate:** 60C continuous discharge for demanding applications.
- **Voltage:** 22.2V (6S configuration).
- **Connector:** Equipped with XT60 plug for wide compatibility.
- **Balance Port:** JST-XH balance port for safe and balanced charging.
- **Durable Construction:** Made with superior lithium polymer raw materials for reliability.



Figure 2.1: Two HRB 6S 4000mAh 60C Lipo Batteries. These batteries are designed for high-performance RC models, featuring XT60 connectors and a compact form factor.



Figure 2.2: Detailed view of the HRB 6S Lipo Battery, highlighting its dimensions (135x42x49mm), 4000mAh capacity, 22.2V voltage, 60C discharge rate, XT60 main connector, and 7-pin balance port. Also shows 10AWG silicone wire for robust power delivery.

3. Specifications

Parameter	Value
Battery Type	Lithium Polymer (LiPo)
Voltage	22.2V
Cell Configuration	6S1P

Parameter	Value
Capacity	4000mAh
Discharge Rate (C)	60C (Continuous) / 120C (Burst)
Main Connector	XT60
Balance Connector	JST-XH (7-pin)
Dimensions (L*W*H)	135 x 42 x 49 mm (5.31 x 1.65 x 1.93 inches)
Weight	590g (1.3 lb / 20.81 oz)

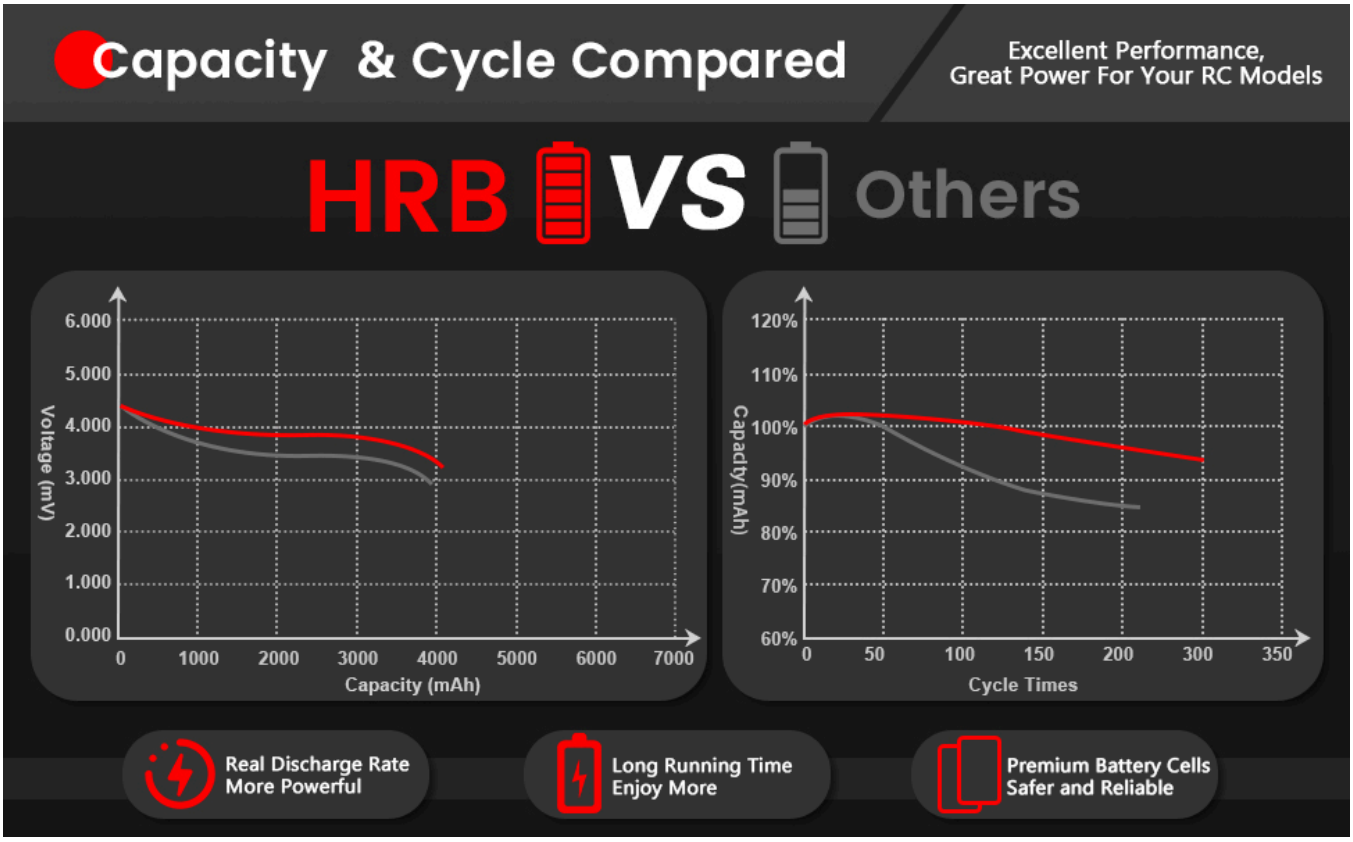


Figure 3.1: Visual representation of the HRB Lipo Battery specifications, including capacity, voltage, discharge rate, configuration, and physical dimensions.

4. Safety Precautions (Important!)

Lithium Polymer (LiPo) batteries are powerful and can be dangerous if not handled correctly. Failure to follow these instructions can result in fire, personal injury, and property damage.

- **Charging:** Always use a LiPo compatible charger with a balance function. Never charge unattended. The safest charge rate for LiPo batteries is 0.5C-1C.
- **Storage:** Store batteries at room temperature (20-25°C / 68-77°F) in a fireproof container or bag, away from flammable materials. Store at a "storage voltage" (typically 3.8V-3.85V per cell).
- **Discharge:** Do not over-discharge batteries below 3.0V per cell. Over-discharging can permanently damage the battery.
- **Physical Damage:** Do not puncture, drop, or crush the battery. If the battery is damaged, swollen, or leaking, discontinue use immediately and dispose of it safely.
- **Temperature:** Do not expose batteries to extreme temperatures (above 60°C / 140°F or below 0°C / 32°F).
- **Disposal:** Dispose of damaged or old batteries responsibly at a designated battery recycling facility. Never dispose of LiPo batteries in regular trash.

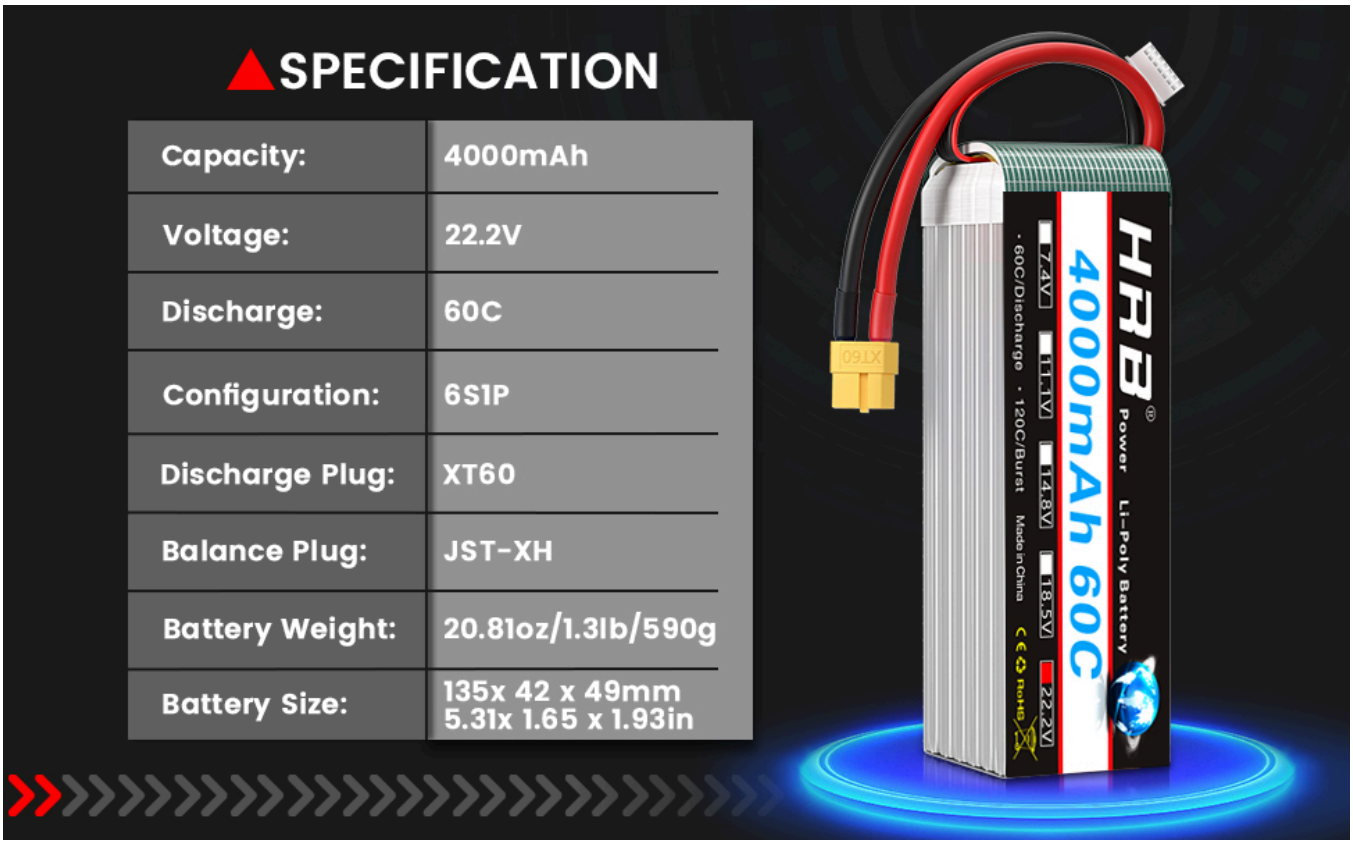


Figure 4.1: Important charging attention for LiPo batteries, emphasizing the safest charge rate of 0.5C-1C and the use of a balance charger.

5. Setup and Initial Use

Before using your HRB Lipo Battery, ensure it is fully charged and inspected for any damage.

1. **Unpacking:** Carefully remove the batteries from their packaging. Inspect for any visible damage such as swelling, punctures, or frayed wires. Do not use if damaged.
2. **Initial Charge:** Connect the battery to a compatible LiPo balance charger. Set the charger to the correct cell count (6S) and capacity (4000mAh). Charge at a rate of 0.5C to 1C (2A to 4A for this 4000mAh battery). Always balance charge.

3. **Compatibility Check:** Verify that your RC model's ESC (Electronic Speed Controller) and motor are compatible with 6S (22.2V) LiPo batteries and can handle the discharge rate. Ensure the XT60 connector is suitable for your model.

6. Operating Instructions

Follow these steps for safe operation of your HRB Lipo Battery in your RC model:

1. **Power Off:** Ensure your RC model's power switch is in the OFF position before connecting the battery.
2. **Battery Placement:** Securely place the battery in the designated battery compartment of your RC model. Ensure it is firmly fastened to prevent movement during operation.
3. **Connect Plugs:** Connect the XT60 main power connector from the battery to the ESC. Ensure the connection is firm and correct.
4. **Balance Lead (Optional for operation, but crucial for charging):** If your model has a balance lead port for real-time monitoring, connect it.
5. **Power On:** Once all connections are secure, turn on your RC model's power switch.
6. **Monitor Voltage:** During operation, monitor the battery voltage. Stop using the battery if any cell drops below 3.5V under load to prevent over-discharge.



Figure 6.1: Visual guide for connecting the HRB Lipo Battery. Steps include turning off power, placing the battery, fastening it, and connecting the plugs correctly.



Figure 6.2: The HRB 6S Lipo Battery installed within an RC car chassis, demonstrating its fit and connection to the vehicle's power system, providing strong power output.

7. Maintenance and Storage

Proper maintenance and storage are crucial for extending the lifespan and ensuring the safety of your LiPo batteries.

- **Regular Inspection:** Before and after each use, inspect the battery for any signs of damage, swelling, or overheating.
- **Storage Voltage:** For long-term storage (more than a few days), charge or discharge the battery to its storage voltage (3.8V-3.85V per cell). Do not store fully charged or fully discharged.

- **Temperature Control:** Store batteries in a cool, dry place, away from direct sunlight, heat sources, and flammable materials.
- **Protective Environment:** Always store LiPo batteries in a fireproof LiPo bag or container.
- **Cleanliness:** Keep connectors clean and free of debris.

8. Compatibility

The HRB 6S 4000mAh 60C Lipo Battery is compatible with a wide range of RC models, including:

- RC Airplanes
- RC Helicopters
- RC Cars
- RC Trucks
- RC Boats

Important: Compatibility is ensured only if the voltage (22.2V), dimensions (135x42x49mm), and plug type (XT60) match the requirements of your specific RC model. Always verify your model's specifications before purchase and use.



Figure 8.1: The HRB Lipo Battery is widely compatible with various RC applications such as airplanes, quadcopters, helicopters, cars, trucks, and boats, provided voltage, size, and plug match.

9. Troubleshooting

If you encounter issues with your HRB Lipo Battery, consider the following common problems and solutions:

Problem	Possible Cause	Solution
Battery not charging	Incorrect charger settings, faulty charger, damaged battery, poor connection.	Verify charger settings (LiPo mode, 6S, 4000mAh). Check all connections. Try another known good charger. If battery is swollen, dispose of it safely.

Problem	Possible Cause	Solution
Battery swells during charge/discharge	Overcharging, over-discharging, physical damage, internal short.	**Immediately disconnect and safely dispose of the battery. Do NOT use a swollen battery.**
Short run time	Battery not fully charged, aging battery, over-discharge, high current draw from model.	Ensure full balance charge. Check battery health with a LiPo checker. Avoid over-discharging. Consider if your model's power demands exceed battery capabilities.
Battery gets hot during use	Excessive current draw, internal resistance, poor ventilation.	Reduce load on the battery. Ensure adequate airflow around the battery in the model. If overheating persists, the battery may be damaged or unsuitable for the application.

10. Warranty and Support

HRB strives to provide high-quality products. For any warranty claims or technical support inquiries, please contact the seller or manufacturer directly through the platform where the product was purchased. Please retain your proof of purchase for warranty validation.
For more information about HRB products, you may visit the official HRB store on Amazon: HRB Store.

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